

EventBusClient

v. 0.1.0

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Chapter 1

Introduction

The component **EventBusClient** provides a modular, high-level messaging framework built on top of the message broker **RabbitMQ**. It abstracts low-level standard protocol operations and adds:

- Plugin support (custom serializers, handlers, message types)
- Dynamic configuration
- Reconnect and lifecycle management

It enables multiple projects to reuse a consistent event bus API without rewriting **RabbitMQ** logic.

The **EventBusClient** is part of a test automation framework called **RobotFramework AIO**, but can also be installed and used stand-alone.

RabbitMQ is an open-source message broker that enables different parts of an application or separate applications to communicate with each other by sending and receiving messages asynchronously. It acts as an intermediary between producers (which send messages) and consumers (which receive and process messages). **RabbitMQ** decouples the sender and receiver, allowing them to operate independently and at their own pace.

Messages are stored in queues until they can be processed, ensuring reliability even if a consumer is temporarily unavailable.

RabbitMQ terms

Term	Description
Producer	Application or service that sends messages
Consumer	Application or service that receives and processes messages
Queue	Buffer that temporarily stores messages
Exchange	Routes messages from producers to queues based on routing rules
Binding	Link between an exchange and a queue, defining how messages are routed
Routing Key	Address-like string used to decide how messages are routed
AMQP	Advanced Message Queuing Protocol, the standard protocol RabbitMQ uses

For more detailed information about these components and frameworks please refer to the corresponding [homepages](#).

Chapter 2

Description

TODO

Chapter 3

EventBusClient.py

3.1 Class: CEventBusClient

Imported by:

```
from EventBusClient.EventBusClient import CEventBusClient
```

TODO

3.1.1 Method: getVersion

Returns the version of EventBusClient as string.

3.1.2 Method: getVersionDate

Returns the version date of EventBusClient as string.

Chapter 4

Glossary

RabbitMQ (open-source message broker)

⇒ [homepage](#)

EventBusClient (high-level messaging framework built on top of **RabbitMQ**)

⇒ [homepage](#)

RobotFramework AIO (**AIO** = **All In One**; test automation framework with extended features, based on the **Robot Framework**)

⇒ [RobotFramework AIO homepage](#)

⇒ [Robot Framework homepage](#)

Chapter 5

Appendix

About this package:

Table 5.1: Package setup

Setup parameter	Value
Name	EventBusClient
Version	0.1.0
Date	09.07.2025
Description	An IPC message-bus based on RabbitMQ message broker
Package URL	python-rabbitmq-messagebus
Author	Nguyen Huynh Tri Cuong
Email	Cuong.NguyenHuynhTri@vn.bosch.com
Language	Programming Language :: Python :: 3
License	License :: OSI Approved :: Apache Software License
OS	Operating System :: OS Independent
Python required	>=3.0
Development status	Development Status :: 4 - Beta
Intended audience	Intended Audience :: Developers
Topic	Topic :: Software Development

Chapter 6

History

0.1.0	05/2022
<i>Initial version</i>	

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